

INTEGRATED AGENTIC MINI-PROJECT

Architectural Report & Antigravity Session Logs

1. Antigravity Session Logs (R2D)

The following logs demonstrate the architectural direction provided to the AI agent to develop the "SME Sentiment & Response Architect" tool.

Architect: "I need to design a tool for SMEs in Almaty. It will be a customer review analyzer. Use Python and Streamlit. The architecture must integrate with the OpenAI API for linguistic processing."

AI Agent: [Generates base application structure and requirements.txt]

Architect: "Add multi-language support. The suggested responses must be generated in both Kazakh and Russian. Also, implement an API key input field in the sidebar to ensure secure deployment in the cloud."

AI Agent: [Updates code, refines system prompts for bilingual output, and adds `st.sidebar.text_input`]

Architect: "Now, implement the Detect to Correct (D2C) error monitoring block. If the API key is missing or the input text is empty, the application should display a user-friendly warning instead of crashing."

AI Agent: [Integrates validation logic using `st.error` and `st.warning` components]

2. IT4IT Product Lifecycle Mapping

Strategy to Portfolio (S2P):

Identified a critical need for SMEs in Almaty to process customer feedback efficiently. An AI-agent approach was selected as the optimal investment to reduce costs compared to hiring dedicated SMM personnel.

Requirement to Deploy (R2D):

The solution architecture utilizes Streamlit for the frontend and OpenAI's LLM for core logic. Automated deployment via Streamlit Cloud ensures high availability (24/7 access).

Request to Fulfill (R2F):

End-users access the tool via a web URL. The consumption model is simplified: review input leads directly to automated analysis and response generation.

Detect to Correct (D2C):

FinOps is managed through OpenAI API usage limits. System reliability is monitored via input validation and error handling to manage hallucinations or missing credentials.

3. Reflective Summary

Managing an AI agent as a Product Architect shifted the focus from syntax to business value and system design. The most significant architectural bottleneck was defining the specific cultural nuances and politeness levels required for the Almaty market, particularly in the Kazakh language. Success depended heavily on prompt engineering to ensure the AI maintained factual consistency without "hallucinating" details not present in the customer reviews.